Mass of the TAL time scalar field: (λ, α_Y) exclusion map and the micron window

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Abstract

We ask what mass the scalar time field of the TAL branch can carry and which masses are already excluded. (i) In vacuum, a mass term $V''(0) = 1/L^2$ cannot be obtained from the branch without a new dimensional constant; we check four routes symbolically. (ii) Without a bare potential the coupling $e^{-\varphi}$ gives a density-dependent mass $m_{\rm eff}^2 = \alpha_Y G\rho e^{-\varphi}/c^2$ (hypothesis H-noV). This mass is far too small for screening ($(R/L_{\rm eff})^2 = 1.66 \times 10^{-10}$ for the Earth), and at $\alpha_Y = 1$ the branch conflicts with the Cassini bound on γ ($1 - \gamma = 0.14742$, ratio $R \approx 5.9 \times 10^3$). (iii) An exclusion map in (λ, α_Y) built from the sub-cm envelope, the AHN 2003 Fig. 4 bound, BREN and Cassini leaves the $\alpha_Y = 1$ branch not excluded only for $2\,\mu\text{m} \le \lambda < 75.877096\,\mu\text{m}$ ($mc^2 \approx 2.60$–$98.7\,\text{meV}$). (iv) In the window $50\,\mu\text{m} \le \lambda < 75.877\,\mu\text{m}$ the drawn bounds would have to tighten by at most a factor 5.26434, falling to 1 at the edge. (v) A reduced source model of the Eöt-Wash 2020 experiment does not reproduce the published bound (0/7 points within $\pm 20\%$); therefore no torque prediction is given. The failure most likely lies in our reduced model (the fitted pendulum offset $r_P = 65.3\,\mu\text{m}$ is far larger than the few-micrometre offsets typical of the experiment). All ratios are conditional ratios to published bounds, not significances; "not excluded" is not confirmation.

1 Scope and reading rules

Every number below is read from a file of the accompanying data package (Section 9); the file is named next to each table or figure. Three categories are used. *Excluded* means that at least one published bound gives $R > 1$. *Not excluded* means $R \le 1$ for every bound that covers the point; this is not a confirmation, and whether an experiment could resolve the effect is not computed. *Not computed* means that no input covers the point; no claim is made there.

- $R = \alpha/\alpha_{\rm lim}(\lambda)$ for laboratory bounds and $R = (1 - \gamma)/b$ for Cassini, with b the bound on $1 - \gamma$. R is a conditional ratio, not a sigma.
- $F = 1/R$ is a bound-tightening factor: the factor by which a published bound would have to improve before it reaches the branch. It is not an experimental sensitivity.

2 Branch and conventions

The TAL branch with Yukawa coupling and its r4 modules (clock, sub-cm, BREN) are documented in the record [6], `https://doi.org/10.5281/zenodo.23960821`. In units $G = c = 1$ the action is

$$S = \int \sqrt{-g}\,\Big[\frac{R}{16\pi} - \frac{1}{2\alpha_Y}\Big((\partial\varphi)^2 + \frac{\varphi^2}{L^2}\Big) - e^{-\varphi}\varepsilon\Big], \tag{1}$$

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with a dimensionless field φ and matter energy density ε. In the weak-field limit the branch gives a Yukawa potential

$$V(r) = -\frac{Gm_1m_2}{r}(1+\alpha e^{-r/\lambda}), \qquad \alpha = \frac{\alpha_Y}{4\pi}, \quad \lambda = L, \quad mc^2 = \frac{\hbar c}{\lambda}, \tag{2}$$

so $\alpha_Y = 1$ means $\alpha_{\mathrm{TAL}} = 1/(4\pi) = 0.0795775$. Constants are CODATA 2018.

3 A vacuum mass cannot be derived

Dimensional lemma. A potential term must have the form $(c^4/G)\,U(\varphi)/\ell^2$ with U dimensionless and ℓ a length. In vacuum the branch contains only G and c, from which no length can be formed; adding $\hbar$ gives only the Planck length. Lengths from the environment are $c/\sqrt{G\rho}$, c/H and $|\mathcal{R}|^{-1/2}$. Hence $V''(0) = 1/L^2 \neq 0$ in vacuum requires a new dimensional constant, which is a hypothesis and not a derivation.

We checked four routes that might seem to avoid this (`theory_checks/checks.py`, output `sympy_checks.json`):

route	symbolic result	vacuum mass
coupling $e^{-\varphi}$ alone	$U'' = 1/L^2 + \alpha_Y\rho e^{-\varphi} + \xi\alpha_Y\rho$	0 for $1/L^2 = 0$; density-set otherwise
non-minimal $\xi\varphi^2\mathcal{R}$	extra $\xi\alpha_Y\rho$ (dust, $\mathcal{R} = 8\pi\rho$)	0 ($\mathcal{R} = 0$ in vacuum)
$f(\varphi)\mathcal{R}$, $F = e^{2\varphi}$	Jordan→Einstein map differs by a total derivative (true); Brans–Dicke form	0 unless a bare potential (a new constant) is added
exponential potential $Ae^{p\varphi}$	minimum with $U'' = q(p+q)\,Be^{-q\varphi_{\min}}$	needs A, a new constant

Kinetic normalization only fixes α_Y and introduces no length. For the cosmological drift we use a linearized test-field solution on a prescribed matter-era GR dust background ($H = 2/3t$, $e^{-\varphi} \approx 1$, negligible scalar backreaction), with the bare mass and the non-minimal term switched off: the solved equation is $\ddot{\varphi} + 2\dot{\varphi}/t = \alpha_Y/(6\pi t^2)$, with solution $\varphi = C_1 + C_2/t + (\alpha_Y/6\pi)\ln t$, i.e. $\varphi \simeq (\alpha_Y/4\pi)\ln a$. Substituting the solution back into this equation gives the residual 0 (`sympy_checks.json`). This is not the full nonlinear solution: restoring $e^{-\varphi}$ in the source of the same background leaves the residual $\alpha_Y/(6\pi t^2)\,[1 - e^{-\varphi}]$, which vanishes only where $e^{-\varphi} = 1$.

Table 1 lists seven symbolic outputs as they are stored; the article build stops if any of them differs from the expression quoted here.

Table 1: Symbolic outputs (`theory_checks/sympy_checks.json`); ξ is the non-minimal coupling, p, q the exponents of the exponential potential.

quantity	stored result
static field equation, $\partial U/\partial\varphi$	$\varphi/L^2 - \alpha_Y\rho e^{-\varphi} + \xi\alpha_Y\rho\varphi$
effective mass, U''	$1/L^2 + \alpha_Y\rho e^{-\varphi} + \xi\alpha_Y\rho$
matter era: linearized test-field solution (prescribed GR dust, $e^{-\varphi} \approx 1$)	$C_1 + C_2/t + (\alpha_Y/6\pi)\ln t$
residual of that solution in $\ddot{\varphi} + 2\dot{\varphi}/t = \alpha_Y/(6\pi t^2)$	0
conformal map differs by a total derivative	true
exponential potential at its minimum, U''	$q(p+q)$ (times $Be^{-q\varphi_{\min}}$)
PPN parameter	$1 - \gamma = 2\alpha_Y/(\alpha_Y + 4\pi)$

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4 Density mass (hypothesis H-noV) and the Cassini conflict

Without a bare potential ($L \to \infty$, hypothesis H-noV) the static equation gives

$$m_{\rm eff}^2 = \frac{\alpha_Y G\rho\, e^{-\varphi}}{c^2}, \qquad L_{\rm eff} = \frac{c}{\sqrt{\alpha_Y G\rho}} \quad (e^{-\varphi} \simeq 1), \tag{3}$$

free of ghosts and tachyons for $\alpha_Y > 0$. It is derived only under H-noV. Table 2 lists the scales for $\alpha_Y = 1$ (`theory_checks/numbers.json`).

Table 2: Density mass scales for $\alpha_Y = 1$ (H-noV).

environment	ρ [kg m^{-3}]	$L_{\rm eff}$ [m]	$L_{\rm eff}$ [AU]	mc^2 [eV]
laboratory (granite)	2700	7.06×10^{11}	4.72	2.79×10^{-19}
Earth (mean)	5513	4.94×10^{11}	3.30	3.99×10^{-19}
Sun (mean)	1410	9.77×10^{11}	6.53	2.02×10^{-19}
interplanetary, 1 AU	8.36×10^{-21}	4.01×10^{23}	2.68×10^{12}	4.92×10^{-31}

Weak screening. Inside a body of radius R, $(R/L_{\rm eff})^2 = (3\alpha_Y/4\pi)\, GM/(Rc^2)$, of the order of the compactness: 1.66×10^{-10} for the Earth and 5.07×10^{-7} for the Sun. Screening therefore needs compact objects; in the laboratory and the solar system the force is effectively long-ranged.

Cassini. With $\alpha = \alpha_Y/(4\pi)$ the Brans–Dicke relation $\gamma = (1-\alpha)/(1+\alpha)$ gives

$$1 - \gamma = \frac{2\alpha_Y}{\alpha_Y + 4\pi}, \qquad 2\omega_{\rm BD} + 3 = \frac{4\pi}{\alpha_Y}, \tag{4}$$

i.e. $1 - \gamma = 0.14742$ at $\alpha_Y = 1$. Bertotti et al. [5] report $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ (`cassini_gamma_input.json`). Taking the 2σ lower edge as a bound on $1-\gamma$, $b = 2.5 \times 10^{-5}$, gives $R = 5.9 \times 10^3$ and $\alpha_Y \le 1.571 \times 10^{-4}$; the 1σ lower edge ($b = 2.0 \times 10^{-6}$) gives $\alpha_Y \le 1.257 \times 10^{-5}$. The H-noV density mass with $\alpha_Y \approx 1$ is therefore in tension with Cassini; the long range of Section 4 is what makes the PPN bound apply.

Inverting the relation for a bound b on $1 - \gamma$ gives

$$\alpha_Y^{\rm max} = \frac{4\pi b}{2 - b}, \tag{5}$$

which is the Cassini entry of the map (Section 5). For the two branch values used in the r4 clock module:

α_Y	$1-\gamma$	R ($b = 2.5 \times 10^{-5}$)	reading
1	0.14742	5.9×10^3	excluded
1/3	0.05168	2.1×10^3	excluded

Both values exceed the bound by more than three orders of magnitude, so the reading does not depend on whether the 1σ or the 2σ edge is used.

5 Exclusion map in (λ, α_Y)

The map (`map_r3/`) combines four bounds on a grid of 400 λ values from 1 nm to c/H_0 and 200 α_Y values from 10^{-6} to 10: (1) the sub-cm envelope of the r4 module (from 1.99986 μm to 1 cm), (2) the AHN 2003 Fig. 4 bound [4] for $\lambda \ge 2$ mm, digitized from the figure (its digitization error relative to the drawn figure is not quantified), (3) the BREN r4 continuation bound for 300 m–50 km, (4) Cassini for $\lambda \ge 1$ AU. For the Cassini layer we use the long-range PPN relation, operationally from $\lambda \ge 1$ AU. We did not compute the finite-λ radio-link response or the error of this approximation;

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this is an assumption of the map layer. As a stability control of the threshold rule only (not a physical validation), moving the threshold from 1 AU to 100 AU changes 0 of the 34 000 cells with $\lambda \geq 1\,\mathrm{AU}$ (`control_cassini_threshold.json`). In each cell $R_{\max}$ over the covering bounds decides the category. The implementation of the AHN bound agrees with the r4 clock module on 201 grid values (maximum relative deviation 2×10^{-16}, `crosscheck_ahn4_impl.json`).

Decision rule. For a bound b that covers λ, with limit $\alpha_{\mathrm{lim},b}(\lambda)$ on the Yukawa strength,

$$R_b(\lambda, \alpha_Y) = \frac{\alpha_Y}{4\pi\,\alpha_{\mathrm{lim},b}(\lambda)}, \qquad R_{\max} = \max_b R_b, \qquad \alpha_Y^{\max}(\lambda) = 4\pi \min_b \alpha_{\mathrm{lim},b}(\lambda), \tag{6}$$

and for Cassini $R = (1-\gamma)/b$ with the α_Y dependence of Section 4. A cell is excluded if $R_{\max} > 1$. The sub-cm envelope is interpolated linearly in $\log\lambda$–$\log\alpha$; the AHN bound is the lower edge of the digitized excluded polygons (95% CL as drawn); BREN takes, at each of its 12 nodes, the least restrictive of its variants and is interpolated log–log between the nodes. Table 3 gives the λ bands and the bounds that cover them.

Table 3: Coverage bands of the grid (`map_r3/map_summary.json`); limits are the first and last grid values of each band.

covering bounds	λ from [m]	λ to [m]
not computed	1.00×10^{-9}	1.81×10^{-6}
sub-cm	2.22×10^{-6}	1.79×10^{-3}
sub-cm + AHN Fig. 4	2.19×10^{-3}	9.06×10^{-3}
AHN Fig. 4	1.11×10^{-2}	2.81×10^{2}
AHN Fig. 4 + BREN	3.44×10^{2}	4.47×10^{4}
AHN Fig. 4	5.47×10^{4}	1.47×10^{11}
AHN Fig. 4 + Cassini	1.80×10^{11}	8.98×10^{14}
Cassini	1.10×10^{15}	1.37×10^{26}

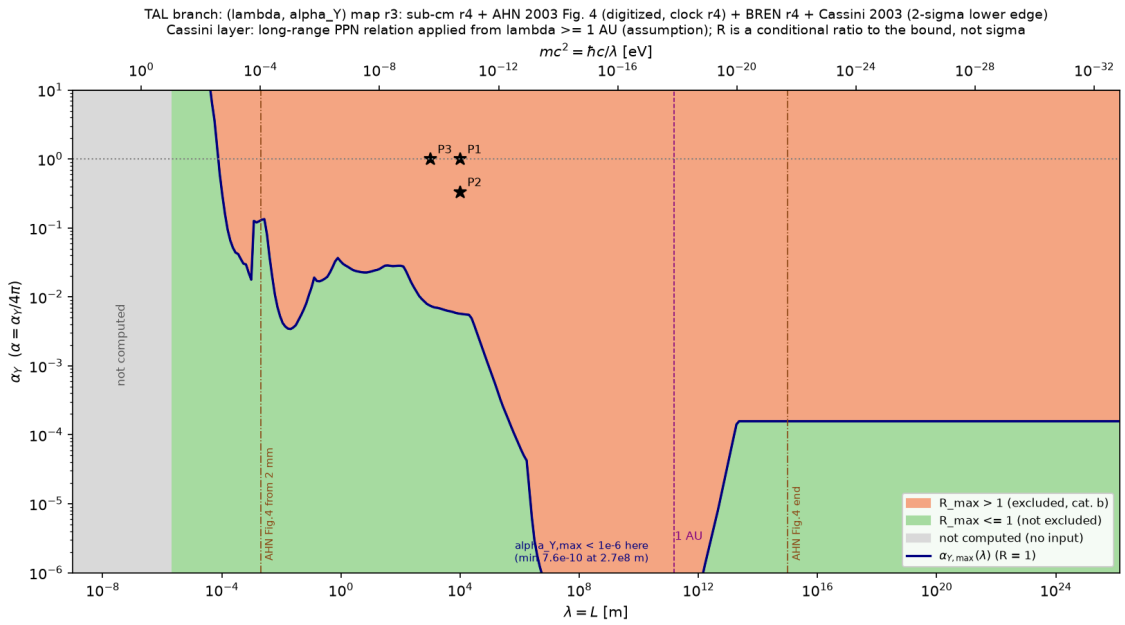


Figure 1: Exclusion map (`map_r3/map_lambda_alphaY_r3.png`). Cell categories: excluded by at least one bound, not excluded (not confirmed), not computed. The dotted grey line marks $\alpha_Y = 1$, the dark-blue curve is $\alpha_Y^{\max}(\lambda)$ ($R = 1$), the dashed purple line marks 1 AU, stars mark P1–P3.

Counts. Of 80 000 cells, 45 817 are excluded, 26 583 not excluded and 7 600 not computed (`map_summary.json`). The not-computed band is the gap from 1 nm to 1.81 μm on the grid; the

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sub-cm envelope starts at 1.99986 μm. As a sensitivity check only, the rule of the r4 clock module (a cell counts as excluded only if every covering bound excludes it) gives 42 830 excluded and 29 570 not excluded cells; the $\alpha_Y = 1$ column below 2 mm and the window edge are unchanged because only the sub-cm envelope covers $\lambda < 2$ mm.

The $\alpha_Y = 1$ branch is excluded for every λ from 75.877096 μm up to c/H_0 without gap and not excluded only for 2 μm $\leq \lambda <$ 75.877096 μm.

Table 4: $\alpha_Y^{\max}(\lambda)$: minimum over the covering bounds; binding bound and other covering bounds (`map_r3/key_values_r3.json`).

λ	mc^2 [eV]	$\alpha_Y^{\max}$	binding	other bounds ($\alpha_Y^{\max}$)
2 μm	9.9×10^{-2}	1.27×10^{7}	sub-cm	–
10 μm	2.0×10^{-2}	38815	sub-cm	–
50 μm	3.9×10^{-3}	5.26	sub-cm	–
75.877 μm	2.6×10^{-3}	1.00	sub-cm	–
100 μm	2.0×10^{-3}	0.342	sub-cm	–
1 mm	2.0×10^{-4}	0.0173	sub-cm	–
1 cm	2.0×10^{-5}	0.00450	AHN Fig. 4	sub-cm: 0.00656
1 m	2.0×10^{-7}	0.0324	AHN Fig. 4	–
100 m	2.0×10^{-9}	0.0283	AHN Fig. 4	–
1 km	2.0×10^{-10}	0.00746	AHN Fig. 4	BREN: 0.120
10 km	2.0×10^{-11}	0.00569	AHN Fig. 4	BREN: 0.104
100 km	2.0×10^{-12}	8.91×10^{-4}	AHN Fig. 4	–
10^8 m	2.0×10^{-15}	2.47×10^{-9}	AHN Fig. 4	–
1 AU	1.3×10^{-18}	3.82×10^{-8}	AHN Fig. 4	Cassini: 1.57×10^{-4}
10^{14} m	2.0×10^{-21}	1.57×10^{-4}	Cassini	AHN Fig. 4: 0.00382
c/H_0	1.4×10^{-33}	1.57×10^{-4}	Cassini	–

The smallest $\alpha_Y^{\max}$ is 7.6×10^{-10} at $\lambda \approx 2.7 \times 10^8$ m (AHN Fig. 4); Cassini binds from $\lambda \approx 2.3 \times 10^{13}$ m. The H-noV scales of Table 2 ($L_{\rm eff} \approx 4.9 \times 10^{11}$–$9.8 \times 10^{11}$ m) fall where $\alpha_Y^{\max} = 1.5 \times 10^{-7}$–$4.1 \times 10^{-7}$; a self-consistent treatment, in which $L_{\rm eff}$ itself depends on α_Y, was not computed.

Reference points of the r4 clock module. For P1 ($L = 10$ km, $\alpha_Y = 1$), P2 ($L = 10$ km, $\alpha_Y = 1/3$) and P3 ($L = 1$ km, $\alpha_Y = 1$) the map reproduces the conservative BREN ratios $R_{\rm cons} = 9.580$, 3.193, 8.355 and the laboratory-compilation ratios $R_{\rm lab} = 175.64$, 58.55, 133.96 (`map_r3/preflight.json`).

6 The micron window

On the $\alpha_Y = 1$ branch, 2 μm $\leq \lambda <$ 75.877096 μm is not excluded by the bounds drawn in Fig. 5 of Lee et al. 2020 [1], digitized in the sub-cm r4 module (`window_signal_r3/`). The envelope over 17 drawn curves is set, from small to large λ, by the unattributed 2–20 μm curve, by Eöt-Wash 2020 [1] from 7.457633 μm and by HUST 2020 [3] from 60.742381 μm. The unattributed curve sets the envelope below 7.457633 μm; it is not the only bound there.

Method. Each drawn curve c gives $\alpha_c(\lambda)$ inside its own range, interpolated piecewise linearly in $\log\lambda$–$\log\alpha$. With

$$\alpha_{\rm env}(\lambda) = \min_c \alpha_c(\lambda), \qquad F_{\rm env}(\lambda) = \frac{\alpha_{\rm env}(\lambda)}{\alpha_{\rm TAL}} = \frac{1}{R_{\rm env}(\lambda)}, \tag{7}$$

the window edge is the λ at which $F_{\rm env} = 1$, found by bisection in $\log\lambda$. The same bisection gives the transitions between the curves that set the envelope and the crossings of $F_{\rm env}$ with 10, 100 and 1000. All reference values (transitions, crossings, $F_{\rm env}$ at 50, 75 and 75.8 μm, the σ-ratio crossing,

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$F_{\mathrm{env}} = 1$ at the edge) agree with an independent re-computation (9/9; tolerance 1×10^{-5} for crossings, 1×10^{-9} for F_{env}; `window_crossings.json`).

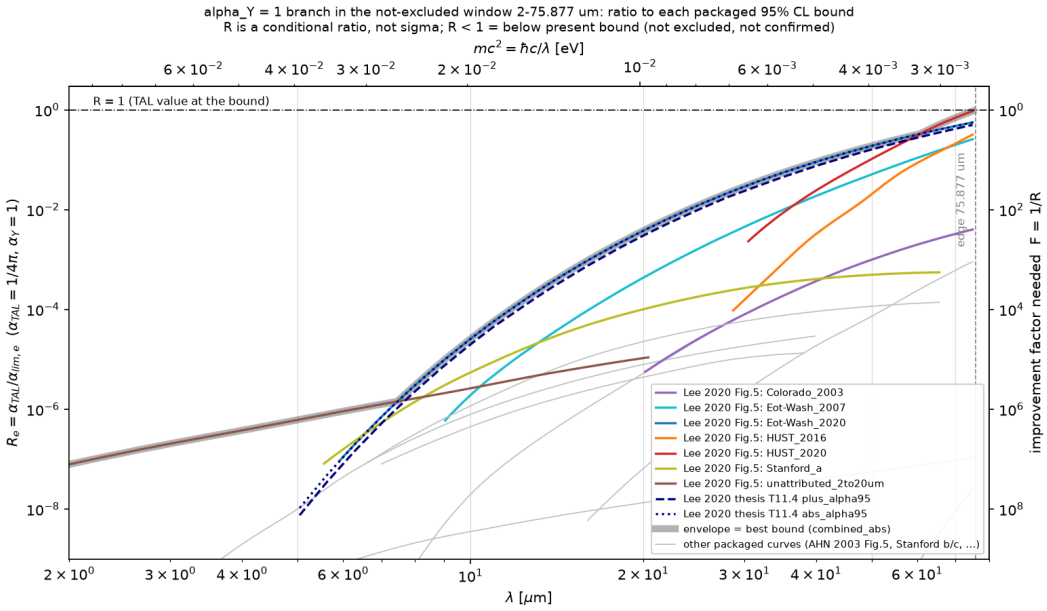


Figure 2: Bound-tightening factor $F_{\mathrm{env}}(\lambda)$ for $\alpha_Y = 1$ (`window_signal_r3/window_ratio.png`). F_{env} is not an experimental sensitivity.

Table 5: Window values for $\alpha_Y = 1$ (`window_key_table.json`). F per bound where it covers λ; thesis column: Table 11.4 $+\alpha_{95}$ of [2].

λ [μm]	mc^2 [meV]	envelope set by	F_{env}	F EW 2020	F HUST 2020	F thesis
2	98.7	unattributed 2–20 μm curve	1.27×10^7	–	–	–
5	39.5	unattributed 2–20 μm curve	1.63×10^6	–	–	1.48×10^8
10	19.7	Eöt-Wash 2020	38815	38815	–	50140
20	9.87	Eöt-Wash 2020	264	264	–	330
30	6.58	Eöt-Wash 2020	34.5	34.5	–	42.3
38.6	5.11	Eöt-Wash 2020	12.5	12.5	53.5	15.3
50	3.95	Eöt-Wash 2020	5.26	5.26	9.64	6.21
70	2.82	HUST 2020	1.43	2.10	1.43	2.39
75	2.63	HUST 2020	1.05	1.78	1.05	2.00

In $50\,\mu\mathrm{m} \le \lambda < 75.877096\,\mu\mathrm{m}$ ($mc^2 \approx 2.60$–3.95 meV) one has $1 < F_{\mathrm{env}} \le 5.26434$, with $F_{\mathrm{env}}(75\,\mu\mathrm{m}) = 1.05316$, $F_{\mathrm{env}}(75.8\,\mu\mathrm{m}) = 1.00454$ and $F_{\mathrm{env}} \to 1$ at the edge. F_{env} crosses 10, 100 and 1000 at 41.093, 23.938 and 16.100 μm (`window_crossings.json`). The envelope equals the minimum of the curves to 2×10^{-16} (relative), and the negative controls pass 6/6: the clean input is accepted and five corrupted inputs are rejected (`test_negative_controls.py`).

Against the thesis table, $\alpha_{\mathrm{TAL}}/\sigma_\alpha = 0.56$, 1.29 and 1.48 at 50, 70 and 75 μm, and $= 1$ at 62.486 μm. These are ratios to a published σ_α, not a predicted significance; a prediction needs a validated source model (Section 7).

7 Source model: an open limitation

To turn the window into a torque prediction we built a reduced Fourier–Bessel model of the Eöt-Wash 2020 pendulum from the open thesis [2] (`source_model_r3/`): 95 runs, $N_{\mathrm{data}} = 285$

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torque values. The prediction is $\tilde{N} = \gamma[R_N N(s+\delta s) + \alpha R_Y Y(s+\delta s)]$ with the empirical off-centre correction R, nuisances γ, δs and a free common pendulum offset r_P. The separation derivative in the χ^2 error term uses the full product rule $\mathrm{d}\tilde{N}/\mathrm{d}s = \gamma[R'_N N + R_N N'] + \alpha\gamma[R'_Y Y + R_Y Y']$; a finite-difference test passes with maximum relative deviation 6×10^{-10} (tolerance 1×10^{-6}, `test_derivative.json`).

Fit statistic. With harmonics h (18ω, 54ω, 120ω), runs j, measured torques $y_{hj} \pm e_{hj}$ and run separation uncertainties δs_j,

$$\chi^2 = \sum_{h,j} \frac{\left(y_{hj} - \tilde{N}_h(s_j)\right)^2}{e_{hj}^2 + \left(\delta s_j \, \mathrm{d}\tilde{N}_h/\mathrm{d}s\right)^2} + \left(\frac{\gamma - \gamma_0}{\sigma_\gamma}\right)^2 + \left(\frac{\delta s}{\sigma_{s_0}}\right)^2. \tag{8}$$

The off-centre factor is $R(s) = c_0 + (1 - c_0)\cos(r_> f)\,(1 - (r_< f)^2/4)$ with $f = c_1 + c_2 e^{c_3 y}$, $r_> = \max(r_P, r_A)$, $r_< = \min(r_P, r_A)$, $y = s$ for Newton and $y = (1 - e^{c_6\lambda})s + c_4 e^{c_5\lambda}$ for Yukawa (s, λ in mm; coefficients from the thesis, `experiment_inputs.json`). Its derivative,

$$\frac{\mathrm{d}R}{\mathrm{d}s} = (1 - c_0)\, f' \Big[- r_> \sin(r_> f)\big(1 - \tfrac{(r_< f)^2}{4}\big) - \cos(r_> f)\, \frac{r_<^2 f}{2} \Big], \qquad f' = c_2 c_3 e^{c_3 y} \frac{\mathrm{d}y}{\mathrm{d}s}, \tag{9}$$

is the term added in source-model version r2 and kept in r3; Table 6 gives the finite-difference test for three cases and the three harmonics. The added term is at most 0.27% of the derivative.

Table 6: Analytic vs finite-difference derivative (`source_model_r3/test_derivative.json`). FD steps: agreement of two step sizes; R' term: largest relative size of the term added in source-model version r2.

case	harm.	max rel. dev.	FD steps	R' term [%]	pass
Newton	120ω	5.1×10^{-10}	1.4×10^{-9}	0.24	yes
Newton	18ω	4.9×10^{-10}	1.6×10^{-9}	0.27	yes
Newton	54ω	5.1×10^{-10}	1.2×10^{-9}	0.12	yes
Yukawa, $\alpha = 1/(4\pi)$, $\lambda = 50\,\mu$m	120ω	5.1×10^{-10}	1.4×10^{-9}	0.23	yes
Yukawa, $\alpha = 1/(4\pi)$, $\lambda = 50\,\mu$m	18ω	4.9×10^{-10}	1.6×10^{-9}	0.26	yes
Yukawa, $\alpha = 1/(4\pi)$, $\lambda = 50\,\mu$m	54ω	5.1×10^{-10}	1.2×10^{-9}	0.12	yes
Yukawa, $\alpha = 1$, $\lambda = 50\,\mu$m, $r_P < r_A$	120ω	5.2×10^{-10}	4.6×10^{-9}	0.0030	yes
Yukawa, $\alpha = 1$, $\lambda = 50\,\mu$m, $r_P < r_A$	18ω	6.0×10^{-10}	4.2×10^{-9}	0.0036	yes
Yukawa, $\alpha = 1$, $\lambda = 50\,\mu$m, $r_P < r_A$	54ω	5.6×10^{-10}	4.6×10^{-9}	0.0016	yes

Newton fit. χ^2 falls from 1236.6 (nominal) to 291.1403 with $r_P = 65.3\,\mu$m, $\gamma = 0.9966$, $\delta s = 0.93\,\mu$m; this agrees with the cross-check value 291.14030018. With three fitted nuisances, two of them with Gaussian priors, the approximate effective number of degrees of freedom is $\nu_{\mathrm{eff}} \approx 284$. The fit is below the pre-declared N_{data}-based benchmark 356.624 and below the ν_{eff} reference 355.498; the benchmark is technical, not a formal χ^2 test.

Validation fails. The rule, fixed before the fit, required the absolute limit $|\alpha_{95}|$ within $\pm 20\%$ of the thesis Table 11.4 at ≥ 2 of 7 λ values. The model gives 0/7, with ratios 1.203–2.118 (Table 7). Hence no TAL torque, signal-to-noise ratio or $\alpha_{\mathrm{TAL}}/\sigma_\alpha$ is derived from this model.

Two α_{95} definitions. The signed $\alpha_{95,\mathrm{pos}}$ and $\alpha_{95,\mathrm{neg}}$ profile thresholds use the fixed $\Delta\chi^2 = 6.17$ rule (the crossings of $\chi^2_{\mathrm{min}} + 6.17$); this value is the fixed comparison recipe of the rule, not a formally established 95% test. The absolute $A = |\alpha_{95}|$ used in Table 7 and in the validation instead satisfies $P(-A \leq \alpha \leq A) = 0.95$ under the Gaussian approximation with mean α^* and σ_α estimated from the $\Delta\chi^2 = 1$ profile width:

$$\Phi\Big(\frac{A - \alpha^*}{\sigma_\alpha}\Big) - \Phi\Big(\frac{-A - \alpha^*}{\sigma_\alpha}\Big) = 0.95. \tag{10}$$

At $50\,\mu$m, for example, $\alpha^* = 0.2959$ and $\sigma_\alpha = 0.1600$ give $A = 0.5591$, while the signed thresholds are $\alpha_{95,\mathrm{pos}} = 0.7011$ and $\alpha_{95,\mathrm{neg}} = -0.0949$ (`model_results.json`).

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Table 7: Validation of the reduced model against the thesis Table 11.4 (`source_model_r3/model_results.json`). $|\alpha_{95}|$ is the absolute limit A with $P(-A \le \alpha \le A) = 0.95$ in the Gaussian approximation (α^*, and σ_α from the $\Delta\chi^2 = 1$ width); it is not one of the signed $\Delta\chi^2 = 6.17$ thresholds.

λ [μm]	σ_α model	σ_α thesis	$\lvert\alpha_{95}\rvert$ model	$\lvert\alpha_{95}\rvert$ thesis	ratio $\lvert\alpha_{95}\rvert$	ratio σ_α	within $\pm$20%
40	0.302	0.286	1.05	0.877	1.20	1.05	no
50	0.160	0.142	0.559	0.419	1.33	1.13	no
63	0.0959	0.0780	0.331	0.218	1.52	1.23	no
71	0.0776	0.0600	0.265	0.162	1.64	1.29	no
79	0.0662	0.0485	0.223	0.126	1.77	1.36	no
100	0.0507	0.0326	0.162	0.0766	2.12	1.56	no
250	0.0254	0.0137	0.0592	0.0290	2.04	1.85	no

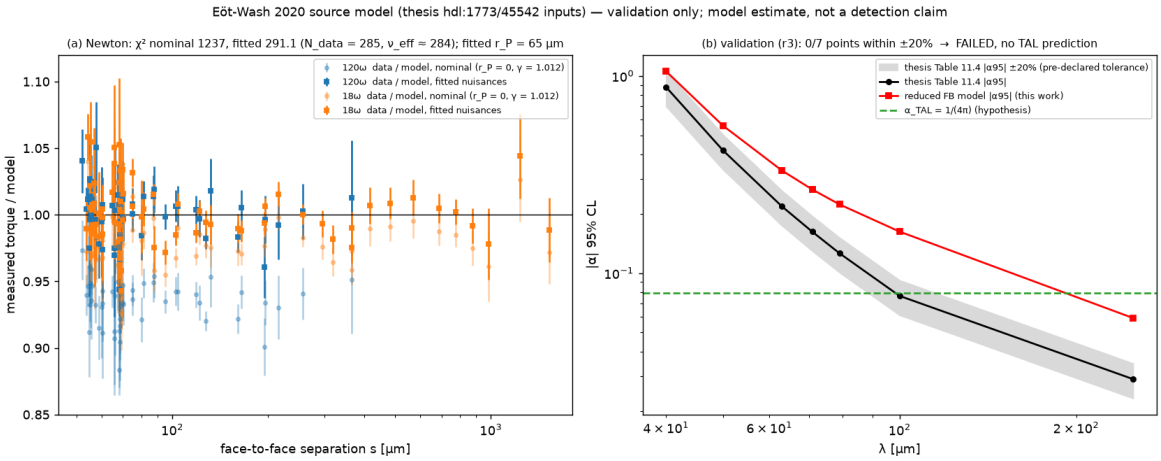


Figure 3: Reduced source model (`source_model_r3/source_model_validation.png`): (a) measured torque divided by the Newtonian model for the 120ω and 18ω harmonics, nominal and with fitted nuisances; (b) $|\alpha_{95}|$ of the model against the thesis Table 11.4 with the $\pm$20% band; the dashed line is α_{TAL} (hypothesis). The validation fails; no TAL prediction is derived.

Sensitivity to the γ prior. Fixing the prior centre at $\gamma_0 = 1.000$ instead of the thesis value gives $\chi^2 = 281.3$ ($\gamma = 0.9950$, $\delta s = 0.26\,\mu$m, $r_P = 67.1\,\mu$m). Changing the prior centre to 1.000 shifts the absolute limits by only 1–3% at the two tested λ values and does not bring them within $\pm$20% (Table 8). This checks one prior variant, not all possible causes of the discrepancy.

Table 8: γ-prior sensitivity (`model_results.json`: validation, sensitivity block); $|\alpha_{95}|$ is the absolute limit A as in Table 7.

λ [μm]	$\lvert\alpha_{95}\rvert$ base	$\lvert\alpha_{95}\rvert$ $\gamma_0 = 1$	$\lvert\alpha_{95}\rvert$ thesis	σ_α base	σ_α $\gamma_0 = 1$	σ_α thesis
50	0.559	0.565	0.419	0.160	0.159	0.142
71	0.265	0.273	0.162	0.0776	0.0774	0.0600

The failure most likely lies in our reduced source model: the fitted $r_P = 65.3\,\mu$m is far larger than the few-micrometre pendulum offsets typical of the experiment, so it is most likely not a physical offset but a sign of an error or a missing ingredient in the model, which can also absorb part of a Yukawa term at larger λ. The per-run pendulum positions are not part of the open data set; this is not treated as the cause of the failure. Reductions of the model that remain to be checked: a single free r_P, measured parameters fixed, ε not fitted, tilt term omitted. Locating the error in the reduced model (for example by reproducing the published constraints from the torques

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of the Supplemental Material of [1]) is the next necessary check step. For HUST 2020 [3] no open full text was available, so no model was built.

8 Limitations

- The vacuum-mass statement is a dimensional and symbolic argument within the branch action; radiative or external constants are outside its scope.
- The matter-era drift is a linearized test-field solution on a prescribed GR dust background ($e^{-\varphi} \approx 1$, no scalar backreaction); the full nonlinear solution was not computed.
- H-noV is a hypothesis; its cosmological drift is not checked against $\dot{G}/G$, BBN or CMB data.
- The AHN and window bounds are digitized from figures; the digitization error is not quantified. BREN is interpolated between 12 nodes.
- For the Cassini layer we use the long-range PPN relation, operationally from $\lambda \geq 1\,\mathrm{AU}$. We did not compute the finite-λ radio-link response or the error of this approximation; this is an assumption of the map layer.
- R and F are conditional ratios to bounds, not sigma and not sensitivities. "Not excluded" is not confirmation.
- The source model did not pass validation; no torque prediction exists. The failure most likely lies in the reduced model itself (Section 7).

9 Data and code availability

The data and code package `TAL_time_field_mass_data_r2.zip` (in this set) contains every script and output used here: `theory_checks/`, `cassini_gamma_input.json`, `map_r3/`, `window_signal_r3/`, `source_model_r3/`, with `README.md`, `LICENSE.txt` and `SHA256SUMS`. Inputs are the clock, sub-cm and BREN r4 zips of `https://doi.org/10.5281/zenodo.23960821`; `extract_r4_inputs.py` verifies their SHA-256 and `run_all.sh` re-runs every module with package-relative paths. No third-party PDF is included.

Reproduction. With the three r4 zips in `inputs_r4/`: `python3 extract_r4_inputs.py` (checks SHA-256 and extracts), then `./run_all.sh`, which runs `theory_checks`, `map_r3`, `window_signal_r3` and `source_model_r3` in this order and stops at the first error. Each script stops with a non-zero exit on a missing input file or key and on a failed preflight; the map and window scripts also stop on a non-finite value in the bounds and curves they read. Before use, the sub-cm envelope is checked by `envelope_guard.py` (identical copies in `map_r3/` and `source_model_r3/`): equal-length one-dimensional arrays, a finite and strictly increasing $\log_{10}\lambda$ grid, no Inf, only positive finite values, no NaN inside the coverage of the packaged curves and no value outside it. Negative controls with one clean and five corrupted envelopes (NaN, Inf, NaN in $\log_{10}\lambda$, 0 and -1 at one covered point) pass 6/6 for the map (`map_r3/test_envelope_guard.json`) and 6/6 for the source-model preflight (`source_model_r3/test_envelope_guard.json`); the window module has its own controls (Section 6). Re-running the value script on the shipped JSON files regenerates the 15 generated article files byte-identically. A fresh full recomputation with `run_all.sh` on the tested software versions (`README.md`) reproduced every numerical output of the previous package version, largest relative deviation 0; on other platforms or library versions, small differences in optimizer results are expected (up to about 10^{-7} relative for a nuisance parameter and below 10^{-9} for $|\alpha_{95}|$ have been observed), within the tolerances of the package tests. In the code, only the input check, labels and file names changed relative to package r1 (map r3, source model r3). The numbers in this article are read from the package outputs by a script; the build stops if a value is missing or not finite, or if the source-model validation were to pass while the text describes a failure.

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Numerical cross-checks with Academia Co-scientist (AI-assisted); not peer review.

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